v158TH FIGHTER WING
HOME OF THE GREEN MOUNTAIN BOYS
THE GREEN MOUNTAIN BOYS
HQ VERMONT AIR NATIONAL GUARD

STRIKE!

Transition to Air Combat Maneuvering

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Air Combat Maneuvers (ACM)

ACM

Air Combat Maneuvers, or ACM, normally involves coordinated maneuvering between two fighters employing BFM to kill, defend, or separate from one or more adversaries Within Visual Range (WVR). The engaged phase can evolve from an intercept phase or an undetected adversary, becoming the highest risk phase of an AA engagement.

Distinct roles, or an **ACM Contract** must be briefed and established among a flight prior to any flight with the potential of an AA engagement in order to ensure effective ACM. This contract will define **engaged, supporting,** and **defensive** roles. Disciplined execution of these roles is key for survivability and lethality within an AA engagement. Any failure in the prosecution of an established ACM contract can lead to disastrous outcomes.

The primary ACM training objectives are:

- **Develop** proficiency in multi-ship coordinated maneuvering.
- **Teach** specific engaged and supporting fighter roles in a WVR fight.
- Developed enhanced **Situational Awareness (SA)**.

Communication

Clear, concise, and correct communications are absolutely **vital** for effective element employment. Each pilot must firmly understand the terms related to each engagement, and be able to use proper terms and communication at the correct moment in an engagement. **Plain english** is perfectly **okay** to replace standard phrasing in any scenario.

However, only **practice** and **critique** will bring proficiency in communication. The time to start improving your communication is not after the first checkin on the radio. The planned cadence, comms procedures, and brevity terms used during the flight should be reviewed in the flight briefing to enhance their effectiveness.

Shortening communications is a bad habit that a pilot will encounter and need to avoid. Nonstandard radio terminology, combined with excessive verbiage can cause confusion, misinterpretation, and override critical information. All missions, regardless of objective, stand as a prime opportunity for all pilots to focus on communications discipline.

Entry

There are two primary entries into an ACM engagement, whether from the vertical, or from the outside of the adversary's turn circle. The supporting fighter can enter from outside an adversary's turn circle, maneuvering in the same direction as the adversary, or enter from the inside of the bandit's turn, maneuvering in the opposite direction. These two are known as **co-flow** and **counter-flow**, and would be entered from the left or right of the adversary and engaged fighter's turn circle, respectively.

In any such case, supporting fighter entry into a fight should happen after an **engaged fighter asks** for assistance, typically through a **"NEUTRAL"** call. From there, the supporting fighter expresses their willingness or ability to engage and gain an offensive posture on the adversary. Four terms are used to indicate roles and responsibilities in an ACM scenario:

- ENGAGED: The fighter with highest contention with the adversary.
- CLEARED: The supporting fighter continues to an offensive posture..
- PRESS: Action is approved, continuation to a supportive posture.
- NEGATIVE: The engaged fighter does not need the support of the other.

Roles

Within an ACM scenario, **two primary roles** exist between fighters facing an adversary: **engaged**, and **supporting**. Again, as you'll see throughout the entirety of this course, **communication** is absolutely key to any engagement. As mentioned on the previous slide, there are four terms used to indicate roles and responsibilities. The **engaged fighter** is maneuvering to **employ** ordnance or tactics to neutralize a threat or adversary, and the **supporting fighter** acts as just that, a **support**, or wingman, if you will.

Role **exchanges** are **not** typically required during an ACM scenario. If the engaged fighter finds itself in an **offensive** posture posed on the adversary, though the fight subsequently becomes **neutral**, the supporting fighter should find it easier to create a Shot of Opportunity (SOO), the original objective of a supporting fighter, by offering it in a supportive manner to the engaged fighter, rather than to PRESS.

However, a role change **should occur** given a scenario in which the supportive fighter is forced into a **defensive** posture.

Bracket & Employment

A standard entry into a high-aspect ACM scenario is seen in the format of a **Bracket**. Herein, the fighter element will create vertical and azimuthal separation, with the goal of decreasing the SA of the adversary or adversaries. Intercepting an adversary with two or more aircraft will frequently overwhelm them.

The bracket also provides the opportunity for roles to be assigned should one fighter be leaned on by the adversary, but also provides a greater opportunity for the supporting fighter to gain an offensive posture once Positive Identification (PID) has been made. A bracket can be formed through an azimuthal separation line-abreast posture of **one mile**, and a vertical separation of **two thousand feet**.

Once PID that the adversary is infact hostile, one fighter should gain a weapons solution through employment of ordnance or tactics.

Egress

Once a kill has been called, an aircraft has been identified as a non-hostile, or the flight should otherwise be recalled from a fight and is not presently engaged, it's key to create not only separation and potential hostile deconfliction, but an easier opportunity to joinup or re-engage should the opportunity or scenario become necessary.

This can typically be achieved through a "SEPARATE" call, followed by a bearing or cardinal direction over the radio from the flight leader.

Example Comms Flow

A typical communications flow looks like this, given a scenario in which PID has failed or a weapons solution has not been cleared.

FL: “Flight, hard right/left, bracket.”
WG: “2.”
FL: “1, bogey leaning on me, engaged.”

[*fighters and adversary/adversaries merge]*

FL: “1, merged, VID hostile, engaged”
WG: “1, 2, PRESS.”
FL: “CLEARED.”
WG: “2.”
FL: “1, FOX 2.”
FL: “1, kill Flanker, left turn, 11,000.”
FL: “Flight, separate, 270.”
WG: “2.”

Additional Study Material

- *High Aspect ACM* by Mike Solyom
- *Multiple Bandits in ACM* by Mike Solyom
- *Air Combat Maneuvering Sortie* by C.W. Lemoine

End

You have completed this lesson portion covering ACM.